

Akylas Stratigakos

UCL Energy Institute, University College London

a.stratigakos@ucl.ac.uk | [Website](#) | [Google Scholar](#) | [ORCID](#)

Research Profile

My research develops machine-learning and optimization methods for reliable, low-carbon energy systems. I focus on decision-aware forecasting, robust and prescriptive analytics, and data-driven tools for power-system and building operations.

Academic Appointments

2025–present	Lecturer (Assistant Professor) <i>UCL Energy Institute, University College London</i> Energy Systems and Data Analytics
2023–2025	Research Associate <i>Department of Electrical and Electronic Engineering, Imperial College London</i> UK-EPICS: Electric Power Innovation for a Carbon-free Society

Selected Publications

- [1] J. G. De la Varga, S. Pineda, **A. Stratigakos**, and J. M. Morales. “End-to-End Pseudo-Measurement Learning for State Estimation under Limited Observability”. In: *Electric Power Systems Research* 265 (2027), p. 114220. [DOI](#).
- [2] G. Van Caelenberg, **A. Stratigakos**, and E. Spyrou. “Constructing Deployment Scenarios for Reserve Deliverability via Adaptive Robust Optimization”. In: *Electric Power Systems Research* 263 (2027). Presented at PSCC 2026, Limassol, Cyprus, 8–12 June 2026, p. 113632. [DOI](#).
- [3] **A. Stratigakos** and P. Andrianesis. “Learning Data-Driven Uncertainty Set Partitions for Robust and Adaptive Energy Forecasting with Missing Data”. In: *IEEE Transactions on Smart Grid* 17.1 (2026), pp. 806–817. [DOI](#).
- [4] **A. Stratigakos**, S. Pineda, and J. M. Morales. “Decision-focused linear pooling for probabilistic forecast combination”. In: *International Journal of Forecasting* 41.3 (2025), pp. 1112–1125. issn: 0169-2070. [DOI](#).
- [5] **A. Stratigakos**, S. Pineda, J. M. Morales, and G. Kariniotakis. “Interpretable Machine Learning for DC Optimal Power Flow With Feasibility Guarantees”. In: *IEEE Transactions on Power Systems* 39.3 (2024), pp. 5126–5137. [DOI](#).
- [6] **A. Stratigakos**, S. Camal, A. Michiorri, and G. Kariniotakis. “Prescriptive trees for integrated forecasting and optimization applied in trading of renewable energy”. In: *IEEE Transactions on Power Systems* 37.6 (2022), pp. 4696–4708. [DOI](#).

Teaching

2026–2027	Data Analytics for Sustainable Buildings (Co-Lead) <i>Sustainable Built Environments, Energy and Resources MEng; 30 credits; 15 students</i> • Co-designing and co-leading the complete module; teaching machine learning, time-series analysis, optimization, and control for smart-building operations.
2025–2027	Energy Data Analytics in the Built Environment (Lead) <i>Energy Systems and Data Analytics MSc; 15 credits; 60+ students</i> • Designed and currently lead the core module, applying machine learning, time-series analysis, optimization, and control across the built environment.

Education

2023	Ph.D., Energy and Processes <i>Mines Paris, PSL University, France</i> <i>Thesis: Towards the Prescriptive Analytics Paradigm for Energy Forecasting and Power System Optimization</i>
2016	MEng, Electrical and Computer Engineering <i>University of Patras, Greece</i> Specialization: Power Systems and Control

Funding and Awards

RESEARCH GRANT	LLMs4Energy: Large Language Models for Energy Systems Modelling — UCL Energy Institute Early Career Research Fund (internal), Principal Investigator, £7.5k (2026)
SELECTED AWARDS	Outstanding Reviewer, IEEE Transactions on Sustainable Energy (2025); Best PhD Thesis Award, Think Smartgrids Association (2024); Best Student Presentation and Travel Grant, International Symposium on Forecasting (2022)

Supervision

PHD	Guillaume Van Caelenberg , Imperial College London, 2023–2026 (co-supervisor); Steven Sperazza , University College London, 2025–2029 (co-supervisor)
DISSERTATIONS	8 MSc and 2 BSc/MEng students supervised

Industry Experience

2022–2023	Volunteer Data Scientist <i>GIVMED, Greece</i> Applied data analytics to redistribute surplus medical supplies to socially vulnerable groups.
2017–2020	Research and Development Engineer <i>Independent Power Transmission Operator, Greece</i> Developed power-system flexibility tools, supported an innovative power-flow-controller installation, and contributed to EU Horizon 2020 projects.

Leadership, Service and Memberships

LEADERSHIP	Vice Chair, CIGRE UK University Hub, London Region (2025–2026); technical programme committee, ACM BuildSys 2026; organizer and chair, invited session on value-oriented forecasting, ISF 2024.
MEMBERSHIPS	Energy Institute, IEEE, CIGRE, ESIG, International Institute of Forecasters
SOFTWARE	Python, MATLAB, R, Julia